

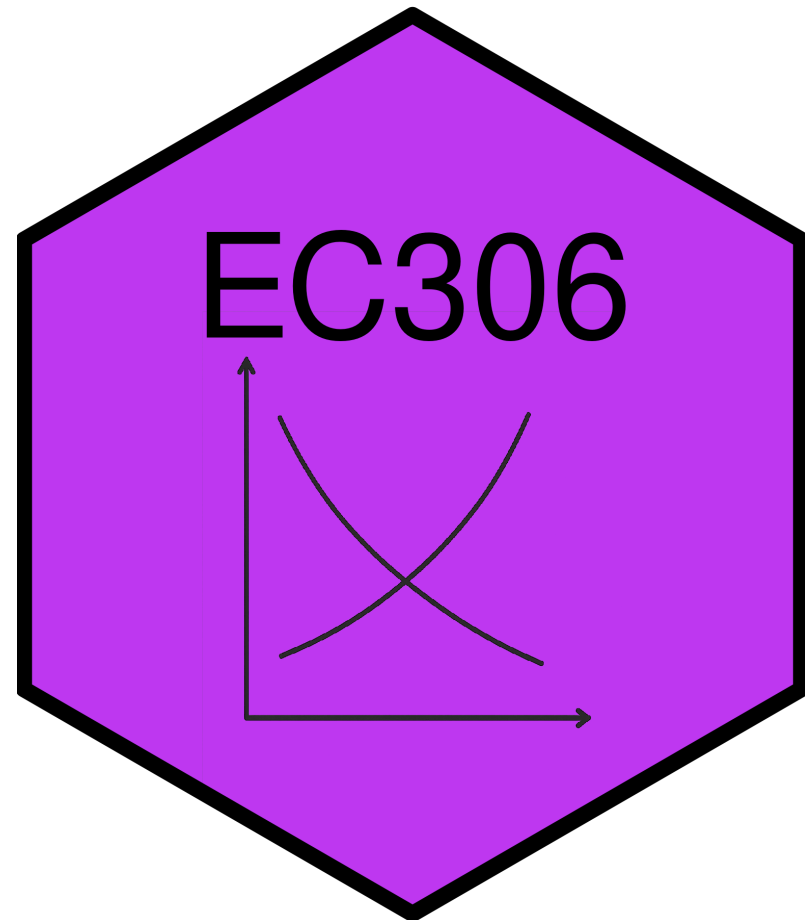
Introduction to the Canadian Labour Market, Measurement, and Research

EC306- Wages and Employment

Justin Smith

Wilfrid Laurier University

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Goals of This Section

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In this section we will:

- Compare labour market outcomes across countries and across Canada
- Review the legal and institutional context in which people and firms decide
- Define the key labour force concepts and the data we measure them with
- See how regression is used, and why correlation is not causation
- Introduce the research designs that let us estimate **causal** effects

Introducing the Canadian Labour Market

Comparing Labour Markets

- Labour economics has a strongly **empirical** tilt
- Comparing outcomes across regions tells us how differences in economic structure or policy matter for people's well-being
- The first thing to describe is how **attached** people are to the labour market
 - Do they participate in paid work, or spend time on unpaid work at home, retirement, volunteering, or school?

Key Definition

The **labour force participation decision** is whether to take part in paid labour market activity. The share of the population doing so is the **participation rate**.

Participation Across Countries

- There is huge variation across countries
 - From about 41% in Jordan to 87% in Iceland
- Canada is relatively high at 80.3% in 2023 — nearly 7 points above the United States
 - These comparisons cover ages 15–64, which is why they run well above the 15-and-over rates in our Canadian charts later
- Most of the cross-country variation comes from differences in women's participation
- Countries with the largest gaps between men and women tend to have lower average incomes
 - Though there are important exceptions

Participation and Economic Development

- Plotting participation against log national income per capita, participation generally rises with income for both men and women
- But there is far more variation among **women** across countries with similar income
 - Mozambique and Madagascar are low-income with very high women's participation
 - The gender gap in Mozambique is only 1.1 percentage points

Note

The reasons are complex. We expect the availability of high-paying opportunities for women, and social norms around women's paid work, to matter a great deal.

Participation and Economic Development

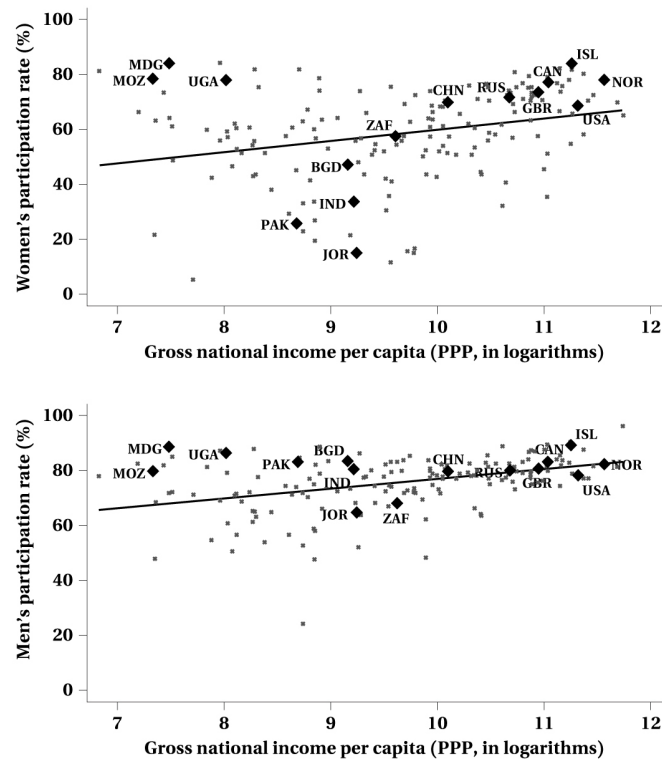


Figure 2.1 Men's and Women's Participation Rates by Level of Economic Development

This figure shows how men's and women's labour force participation varies with the level of economic development as measured by per capita gross national income in 2023. Each point indicates a country's participation rate (with select country abbreviations in Table 2.1) and the natural logarithm of gross national income per capita. The "PPP" (purchasing power parity) is an adjusted measure of per capita national income that accounts for differences in the cost of living between countries. The estimated regression line relates the participation rate to log per capita income. Both panels show that the participation rate generally rises with national income, but there is more variation in the participation rates of women across countries with similar income.

Source: World Bank, World Development Indicators, <https://databank.worldbank.org/source/world-development-indicators#>, series NY.GNP.PCAP.PP.CD, accessed February 11, 2025.

Differences Within Canada

- Nearly 21 million people were employed in Canada in 2024, unevenly distributed
 - Ontario largest at about 8 million; Prince Edward Island smallest at 93,000
- Most people work in [service](#) industries — about 80% nationally
 - Alberta is lower, because resource industries are a larger share of jobs
- Populations differ too: median age is highest in Newfoundland and Labrador, lowest in Manitoba

Both labour demand and supply differ by province, given who lives there.

Two More Indicators

Key Definitions

The **job vacancy rate** is the number of jobs that need to be filled, as a percentage of all jobs — occupied or vacant. It was 3.2% nationally in 2024, highest in British Columbia.

The **union differential** is the percentage difference between average weekly earnings of union and non-union employees. It was 7.3% nationally, but varies widely — over 30% in Prince Edward Island.

- We return to vacancies in Chapter 8 and to unions later in this section

iClicker Question

Across countries, most of the variation in overall labour force participation comes from differences in:

- A) Men's participation
- B) Women's participation
- C) Retirement ages
- D) Unemployment rates
- E) Population size

Institutional and Legal Context

Introduction

- Before analyzing any market, we need the **context** in which people and firms decide
- Institutions and legislation shape the employer–employee relationship
- Rules do not just constrain behaviour — they change **wages and employment**

Canada and Indigenous Peoples

- Inuit, First Nations, and Métis Peoples have a special constitutional relationship with the Crown
 - Including “existing aboriginal and treaty rights” under [section 35 of the *Constitution Act, 1982*](#)
- Laws and policies are being reviewed against the *United Nations Declaration on the Rights of Indigenous Peoples*



Tip

We look at contemporary effects of Canada’s historical policies in Chapter 15.

Canada and Indigenous Peoples

Which rules apply at work?

- Employment relationships follow the same legislation as everyone else, depending on **industry**
 - Federal jurisdiction covers band councils and Indigenous self-governments

Taxation:

- Employment income may be tax exempt when the work is substantially “situated on a reserve” and done by a registered First Nations individual
 - Section 87 of the *Indian Act*

Federal and Provincial Jurisdiction

- The *British North America Act* of 1867 divided power between the two levels of government
- Which government regulates your job depends on [what industry you work in](#), not where you live

[Federal jurisdiction](#) covers the federal public service plus a short list of industries:

- Interprovincial and international transportation
- Telecommunications and broadcasting
- Banks — but **not** credit unions
- Postal and courier services

The provinces have everything else — the large majority of workers.

The Canada Labour Code

Key Definition

The [Canada Labour Code \(CLC\)](#) sets labour standards for industries under federal responsibility. It has four parts.

- Part I — Industrial Relations: collective bargaining and unionization
- Part II — Occupational Health and Safety
- Part III — Labour Standards: employment conditions and protections
- Part IV — Penalties and procedures for noncompliant employers

Not every part applies to every worker: federal private sector gets all four, the federal public sector only II and IV.

Provincial Employment Standards

- Each province has an *Employment Standards Act* or *Code* setting rules on:
 - Minimum wages, vacation time, parental and sick leave
 - Termination of employment, limits on hours
- We care because these rules change the decisions of workers and firms — and therefore *wages and employment*

Minimum Wages in Canada, September 2026

Jurisdiction	Wage	Jurisdiction	Wage
Canada (federal)	\$18.15	Manitoba	\$16.00
Newfoundland and Labrador	\$16.35	Saskatchewan	\$15.35
Prince Edward Island	\$17.00	Alberta	\$15.00
Nova Scotia	\$16.75	British Columbia	\$18.25
New Brunswick	\$15.90	Nunavut	\$20.17
Quebec	\$16.60	Northwest Territories	\$17.20
Ontario	\$17.60	Yukon	\$18.51

- Most jurisdictions now [index annually](#) — Ontario, Manitoba, Nova Scotia, and PEI all adjust on [October 1, 2026](#), partway through this course
- Alberta has not changed since 2019, the longest freeze in the country

 We analyze minimum wages properly in Chapter 12.

Labour Relations, Equity, and Human Rights

- A separate provincial *Labour Relations Act* governs forming a union, negotiations, and dispute resolution
 - This helps explain differences across provinces in unionization rates

! Employment Equity vs. Pay Equity

Employment equity removes barriers to employment for designated groups — women, Indigenous Peoples, persons with disabilities, and members of visible minorities.

Pay equity requires that women and men receive the same pay for work of **equal value**.

- The *Canadian Human Rights Act* protects against discrimination on grounds including race, religion, age, sex, sexual orientation, gender identity, disability, and family status
- Provinces have related legislation for workers outside the federal jurisdiction

iClicker Question

A teller at a credit union in Ontario is covered by:

- A) The Canada Labour Code, because banking is federal
- B) Ontario's employment standards legislation, because credit unions are provincial
- C) Both, equally
- D) Neither
- E) Whichever the employer chooses to register under

Quantifying the Canadian Labour Market

Types of Data

Labour economists catalogue data by the [unit of observation](#).

- Aggregate (time-series) — economy-wide measures like the unemployment rate, inflation, or GDP, reported over time
- Cross-section microdata — earnings, hours, or education at the [individual level](#) at one point in time
- Panel (longitudinal) — the *same* individuals followed over several periods

Why Panel Data Matter

- Panel data let us study **dynamics** — transitions into and out of the labour market
- They also let us control for things we normally cannot observe but that are **fixed** within a person, like motivation
- Few Canadian surveys are longitudinal, and access is restricted for privacy
 - The Longitudinal Administrative Database follows income through tax records
 - The Canadian Employer–Employee Dynamics Database links firms to their workers

Note

Increasingly researchers combine sources, including “big data” like online job ads and text analysis of media reports.

Where the Numbers Come From

- The [Labour Force Survey \(LFS\)](#) measures all of the concepts above
 - Monthly, about 68,000 households and 100,000 individuals aged 15+
 - Its main purpose is the official unemployment rate used to administer EI
 - Coverage excludes reserves and extremely remote areas
- The [Census](#) runs every five years — most recently in May 2026
 - Adds characteristics the LFS does not have: ethnicity, language, country of birth
- The [Canadian Income Survey \(CIS\)](#) is annual, with much more detail on earnings and family circumstances

Where the Numbers Come From

- Two employer-side surveys matter too:
 - SEPH — jobs, earnings, and hours within industries
 - Job Vacancy and Wage Survey — vacancies by occupation



Finding the data yourself

Three routes are open to students: [Statistics Canada](#) PUMFs, the [Data Liberation Initiative](#) through your library, and the [Research Data Centres](#) for confidential files.

We use the CIS later in this section, and you will work with this kind of data yourself.

Key Definitions: The Labour Force

! Important

Labour Force (LF): people in the eligible population who participate in labour market activities, as either employed or unemployed

Labour Force Participation Rate (LFPR): the share of the eligible population in the labour force

$$LFPR = LF / POP$$

Eligible population (POP)

- Aged 15 and over
- Non-institutionalized
- Not living on reserve
- Civilians

Not in the labour force

- Retired people
- Unpaid work in the home
- Full-time students
- People who gave up looking

Key Definitions: Employment and Unemployment

People in the labour force are either **employed** or **unemployed**.

! Important

Employed (E): worked during the reference period — including those normally employed but temporarily absent

Unemployed (U): did not work, but were available and actively looked in the past four weeks, are waiting to start a job within four weeks, or are on temporary layoff expecting recall

$$ER = E/POP \quad UR = U/LF$$

- Economists split the choice in two: the **extensive margin** is whether to work at all, the **intensive margin** is how many hours once you do
 - We measure the intensive margin with hours worked — full time or part time

The Labour Force Hierarchy

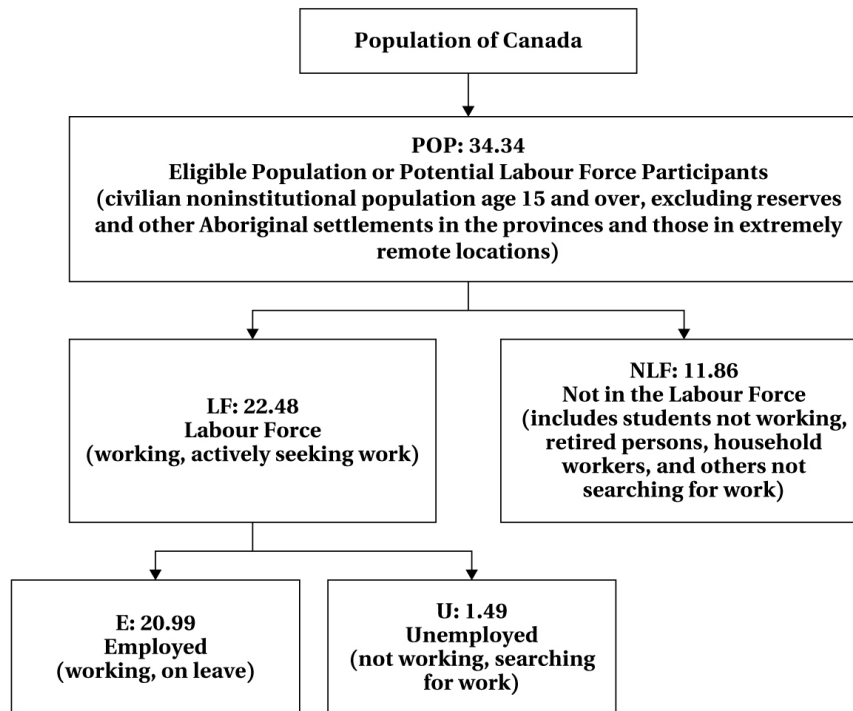


Figure 2.2 Labour Force Concepts (numbers as of January 2025)

The labour force is the sum of those individuals either working (employed) or not working but searching for work (unemployed). The unemployment rate is the percentage of the labour force that is unemployed, while the employment rate is the percentage of the eligible population that is employed.

Notes:

1. Population figures in millions.
2. Labour force participation rate: $LFPR = LF / POP = 22.48 / 34.34 = 65.5 \%$
3. Unemployment rate: $UR = U / LF = 1.49 / 22.48 = 6.6 \%$
4. Employment rate: $ER = E / POP = 20.99 / 34.34 = 61.1 \%$

Source: Adapted from the Statistics Canada Table 14-10-0287-02 (Labour Force Characteristics by Age Group, Monthly, Seasonally Adjusted), accessed February 21, 2025.

iClicker Question

A 22-year-old full-time student who is not looking for work is counted as:

- A) Employed
- B) Unemployed
- C) In the labour force, but not employed
- D) Not in the labour force
- E) Outside the eligible population

iClicker Question

In January 2025 the eligible population was 34.34 million, the labour force 22.48 million, employment 20.99 million, and unemployment 1.49 million. What was the unemployment rate?

- A) 4.3%
- B) 6.6%
- C) 61.1%
- D) 65.5%
- E) 93.4%

What the Data Look Like

With the concepts and the sources in hand, let's look at real Canadian data.

- First the Labour Force Survey — how participation, employment, unemployment, and hours have moved over time
- Then the Canadian Income Survey — where income comes from, and how earnings differ across people

Note

None of this is causal — these are [descriptions](#). They are the facts the rest of the course sets out to explain.

A Century of Participation

- The census lets us push the participation story back to 1901
- Men's participation has drifted down the whole century — from just over 90% to about 70% by 2024
- Women's participation was below 30% as recently as 1961, then rose sharply before levelling out in the 2000s at around 61%
- The rise of women's participation is one of the biggest changes in the Canadian labour market on record
 - Candidate explanations — wages, education, contraception, social attitudes, life expectancy — are questions this course comes back to

A Century of Participation

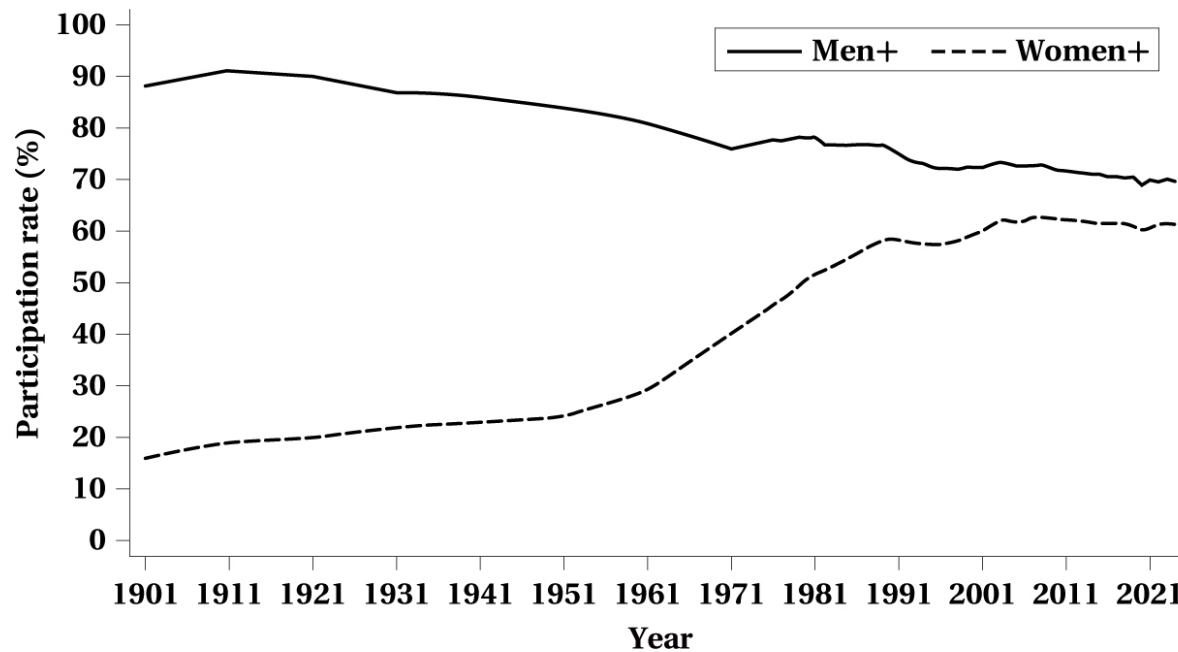
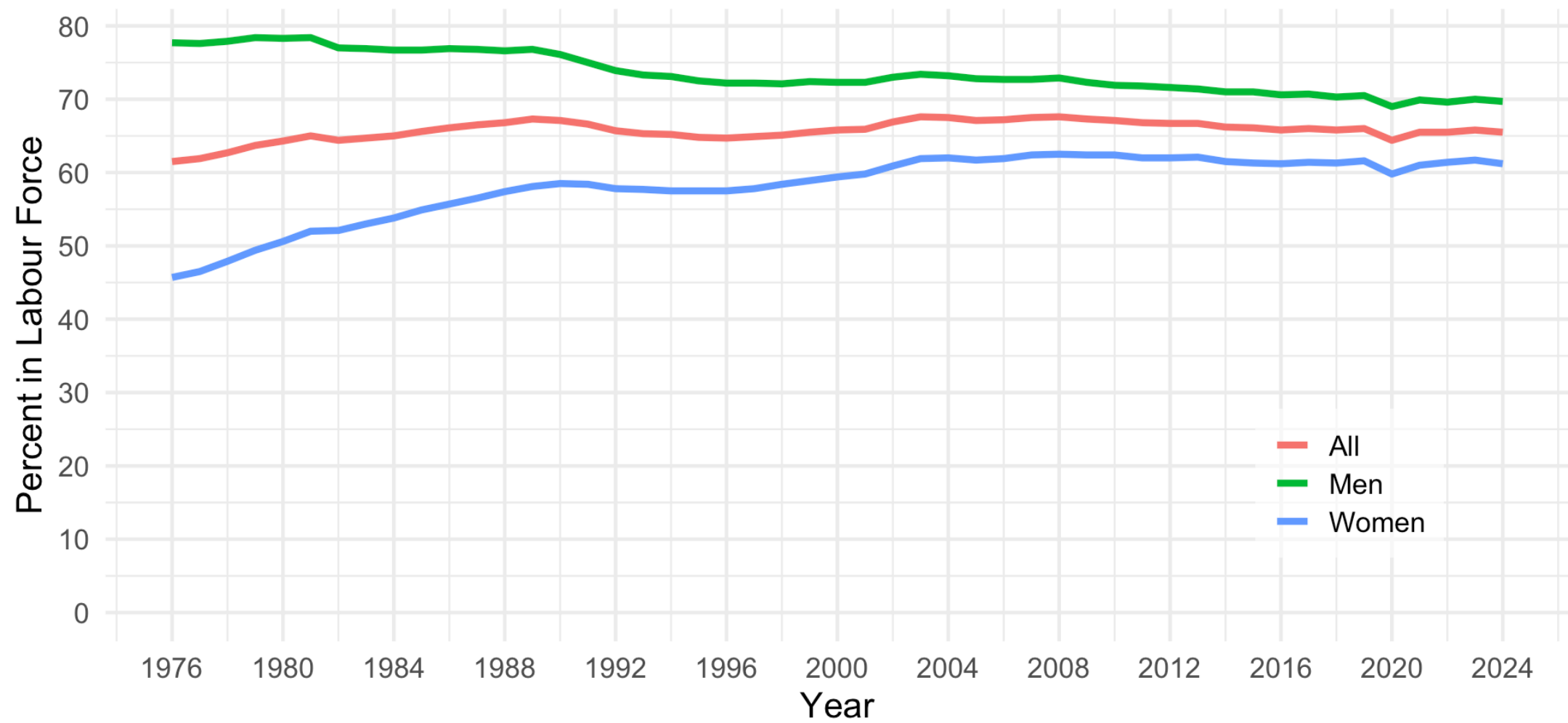


Figure 2.3 Labour Force Participation Rates by Gender, Canada, 1901–2024

Labour force participation rates for men and women aged 15 and over are plotted by Census years, from 1901–1971, and then each year from 1976–2024. The general trends suggest a gradual reduction of labour force attachment for men, in contrast to a sharp increase for women beginning in the 1950s before levelling out in the 2000s. Note that the LFS changed its questionnaire in 2022 to allow respondents to report their nonbinary gender.

Sources: Data adapted from M. Gunderson, “Work Patterns,” in *Opportunity for Choice: A Goal for Women in Canada*, ed. G. Cook (Ottawa: Statistics Canada, 1976), p. 97 for 1901–1971; Participation rates for 1976–2024 are from Statistics Canada Table 14-10-0327-01, accessed February 24, 2025. Statistics Canada provides detailed information about questionnaire changes on its website.

Labour Force Participation Rate, Canada



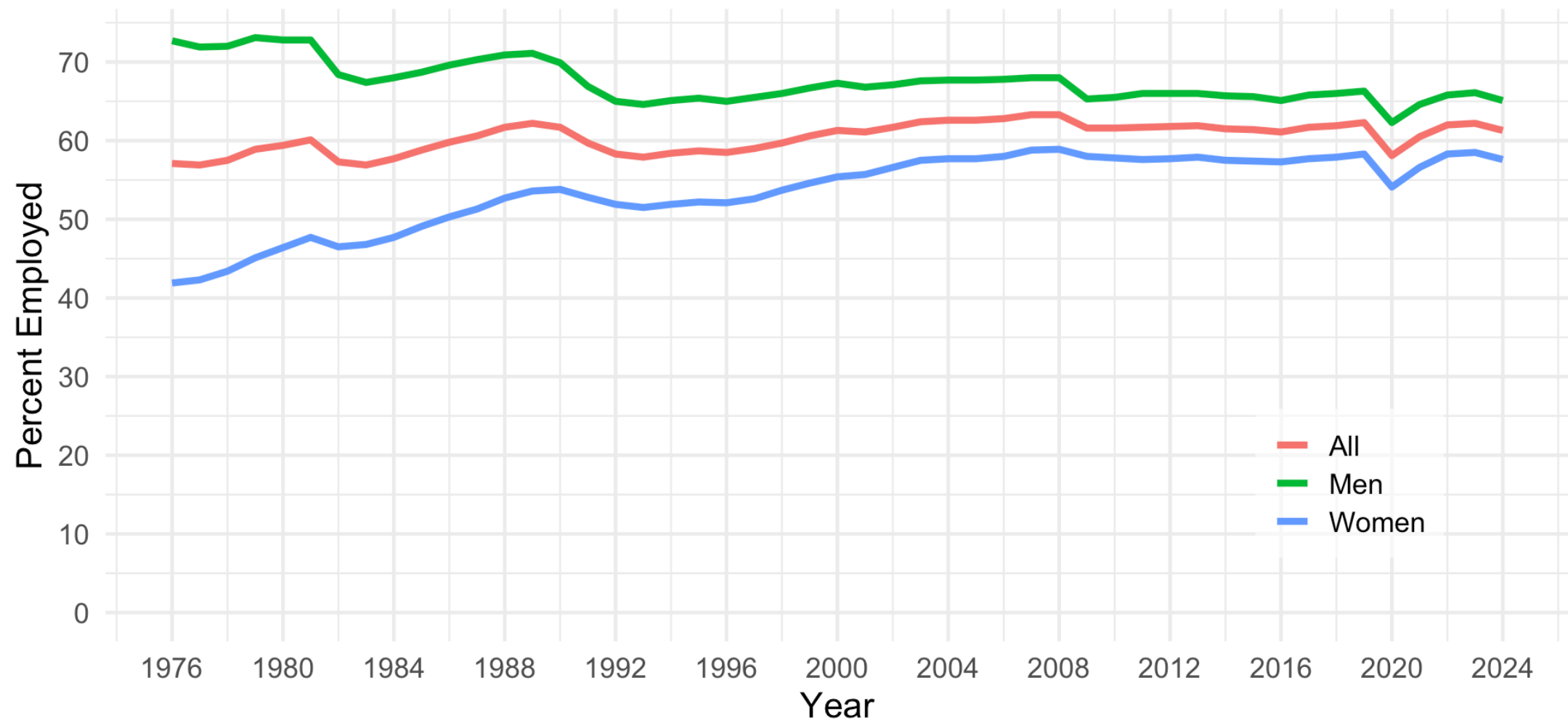
Source: CANSIM Table 14-01-0327-02

iClicker Question

Which statistic is the best proxy for the **demand** side of the labour market?

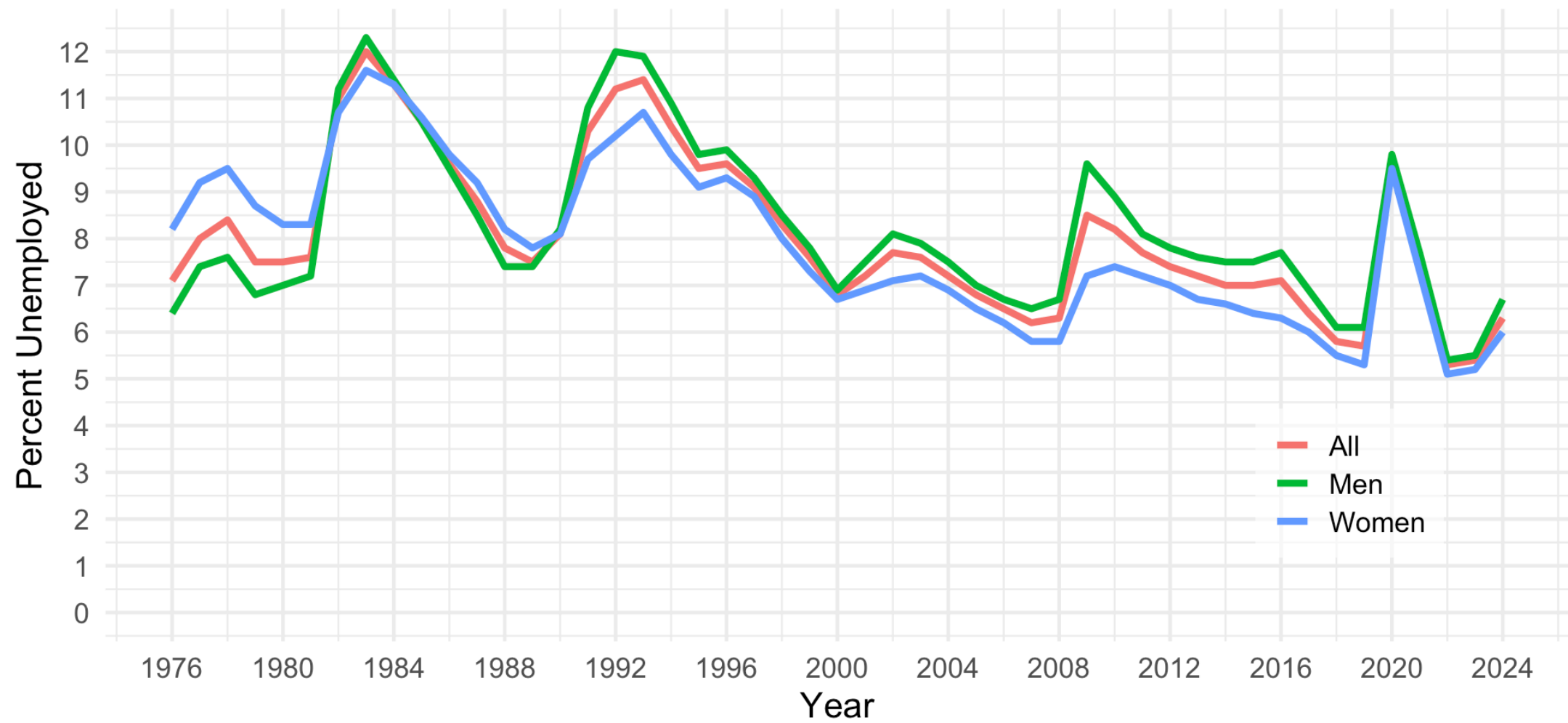
- A) The participation rate
- B) The job vacancy rate
- C) The employment rate
- D) The unemployment rate
- E) The size of the eligible population

Employment Rate, Canada



Source: CANSIM Table 14-01-0327-02

Unemployment Rate, Canada



Source: CANSIM Table 14-01-0327-02

iClicker Question

Someone who has been unemployed for a year stops looking for work. Holding everything else constant, what happens to the measured statistics?

- A) The unemployment rate rises; the participation rate is unchanged
- B) The unemployment rate falls; the participation rate falls
- C) The unemployment rate falls; the participation rate is unchanged
- D) Neither changes, because the person is still jobless
- E) The unemployment rate rises; the participation rate falls

Discouraged Workers and Hidden Unemployment

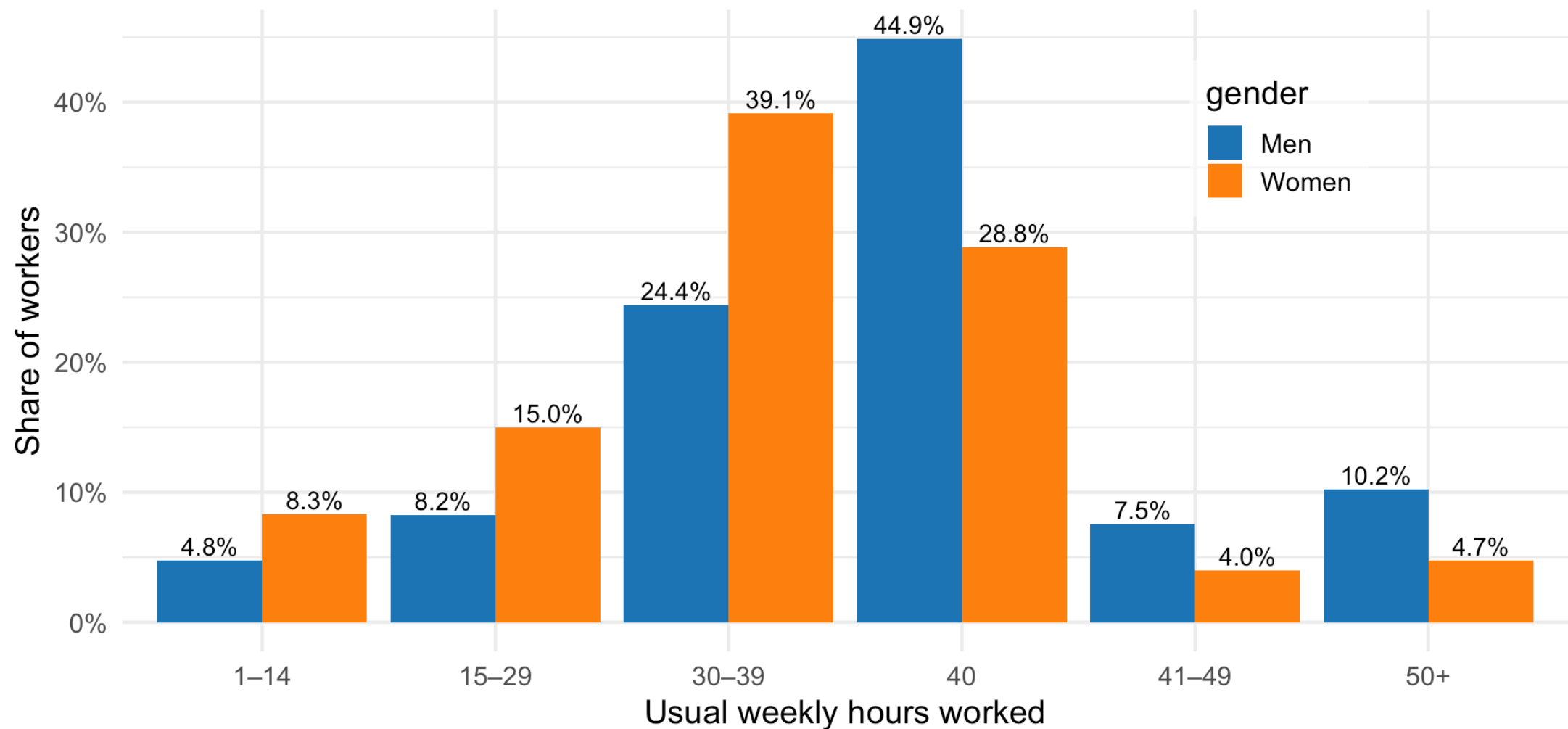
! Key Definitions

Discouraged workers: people who want a job but have stopped searching because they believe no work is available

Hidden unemployment: joblessness the unemployment rate misses, because non-searchers are counted as outside the labour force

- The previous question is exactly this mechanism — discouragement makes the unemployment rate **look better** as the labour market gets worse
- People who want work but are not searching have **marginal labour force attachment**
 - Jones and Riddell (1999) find they behave more like the **unemployed** than like others outside the labour force
- So “unemployed” versus “not in the labour force” is a **measurement line**, not a bright line in behaviour

Usual Weekly Hours: Men vs Women

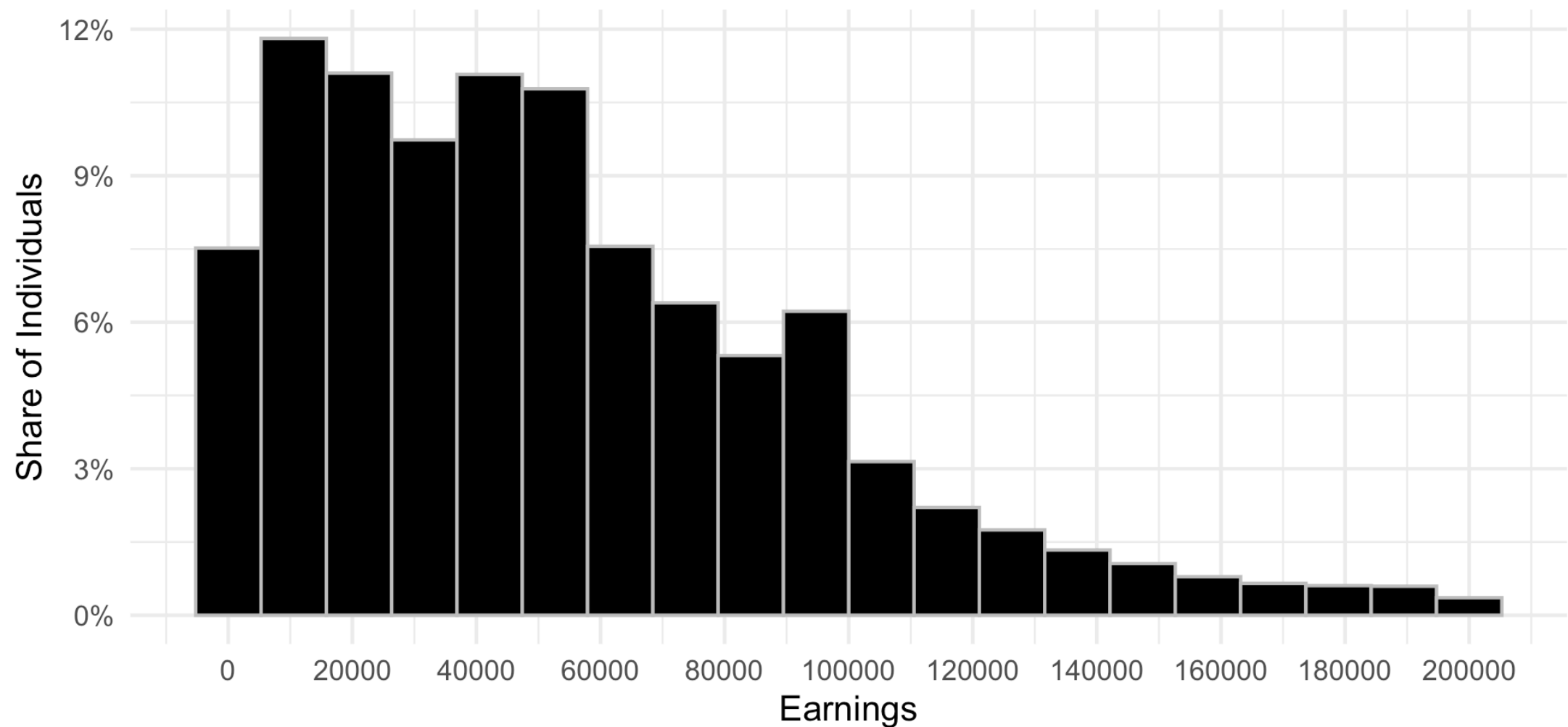


Source: Labour Force Survey April 2025

Income Sources

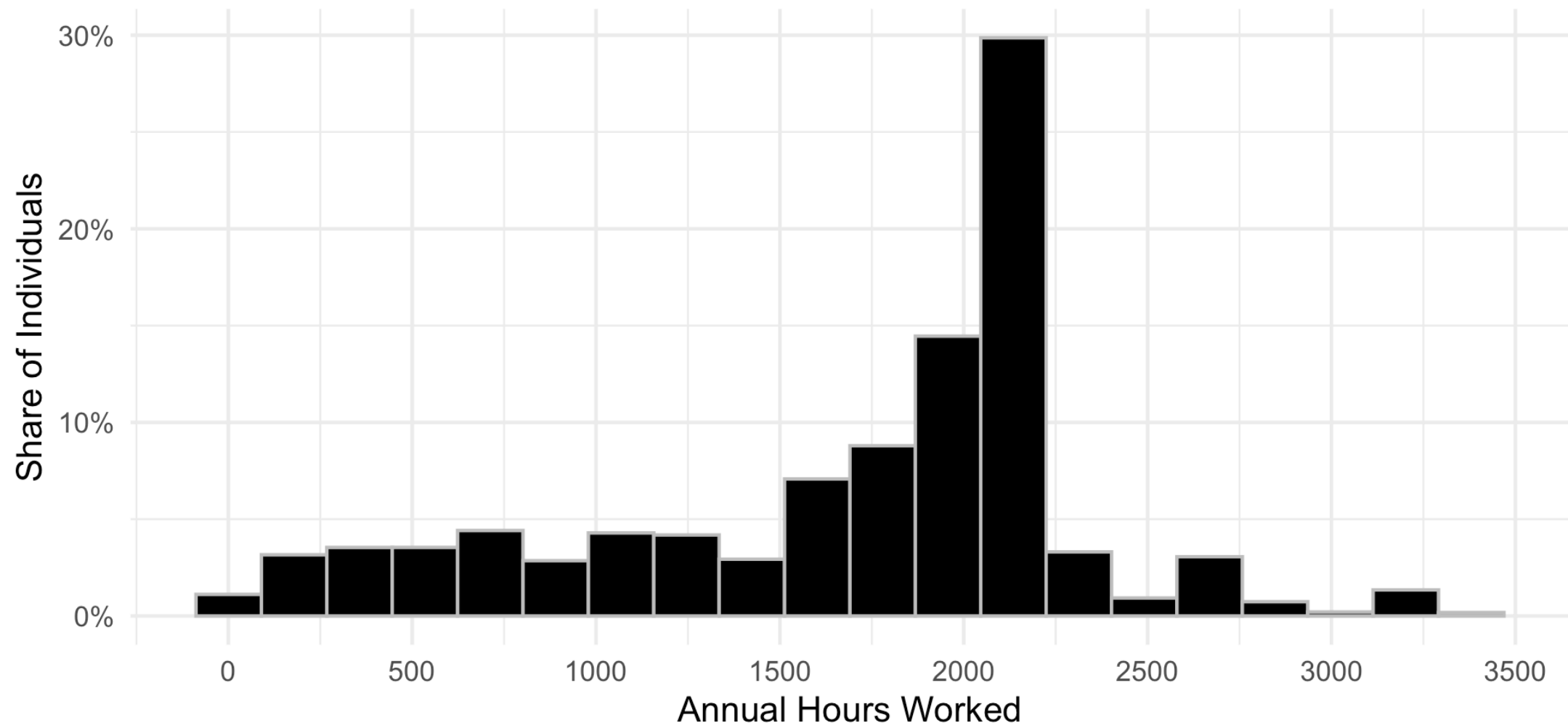
Variable	All		Men		Women	
	Mean	% Positive	Mean	% Positive	Mean	% Positive
Earnings	50,326	84.0	60,216	86.6	40,408	81.5
Investment Income	2,706	28.5	3,291	28.6	2,120	28.3
Pension	1,561	8.2	1,556	7.3	1,567	9.1
Other Income	2,069	17.7	1,926	16.0	2,212	19.4
Government Transfers	4,671	82.6	3,252	78.0	6,094	87.2
Income Taxes	11,012	72.8	14,191	77.2	7,824	68.5
Annual Hours Worked	1,385	82.0	1,546	85.6	1,223	78.4
Wage	40	75.6	44	78.4	36	72.9

Distribution of Earnings



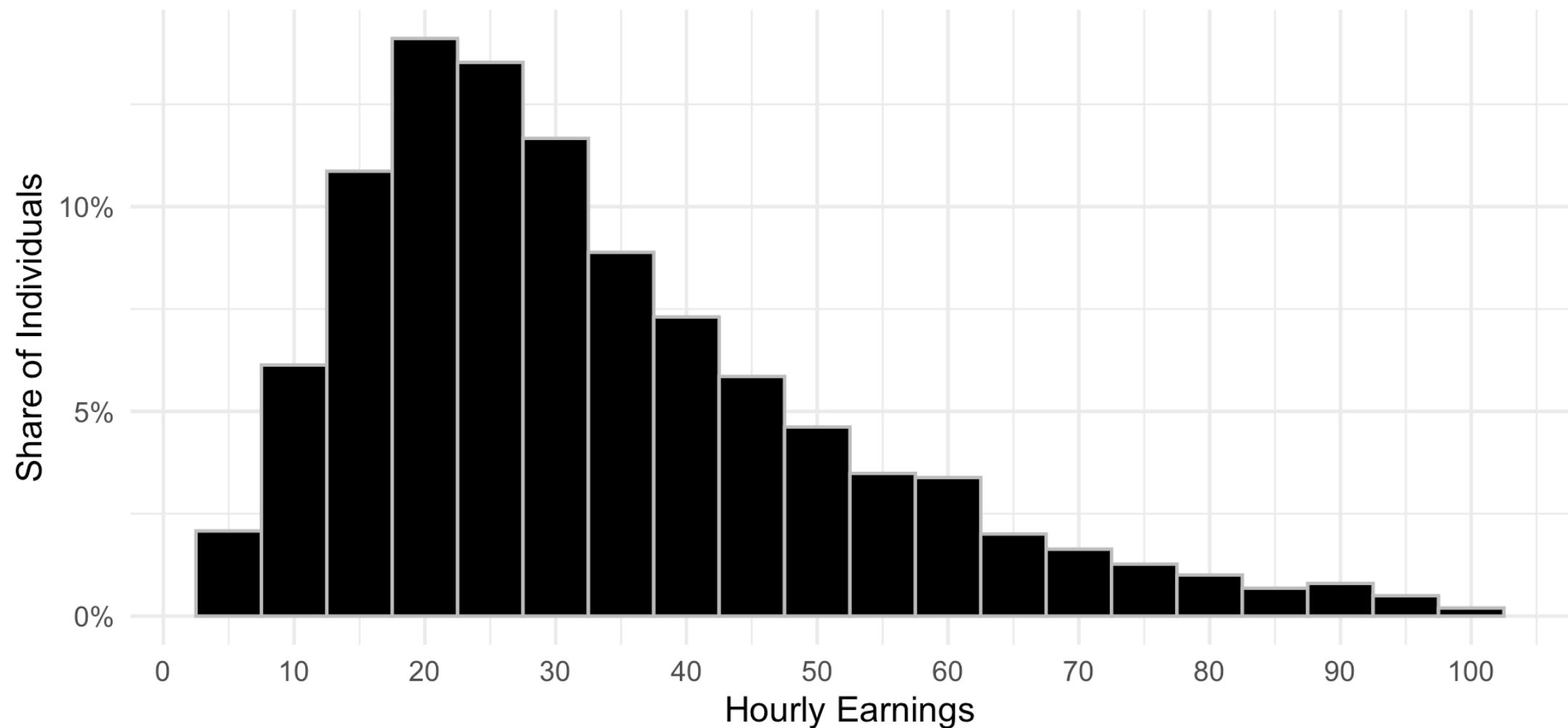
Source: Canadian Income Survey (2022). Plot includes earnings from the 1st percentile to the 98th percentile

Distribution of Annual Hours Worked



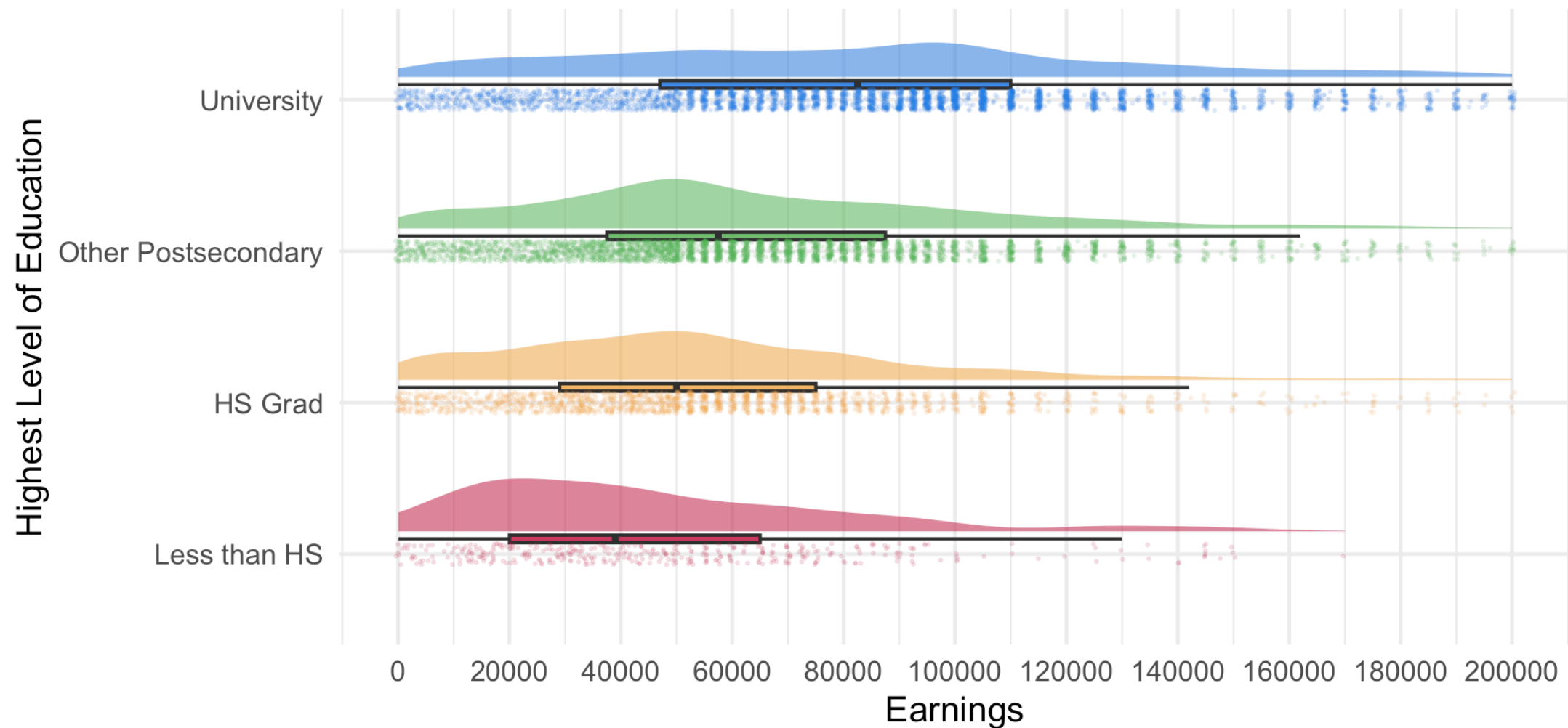
Source: Canadian Income Survey (2022). Plot includes hours from the 1st percentile to the 99th percentile

Distribution of Hourly Earnings



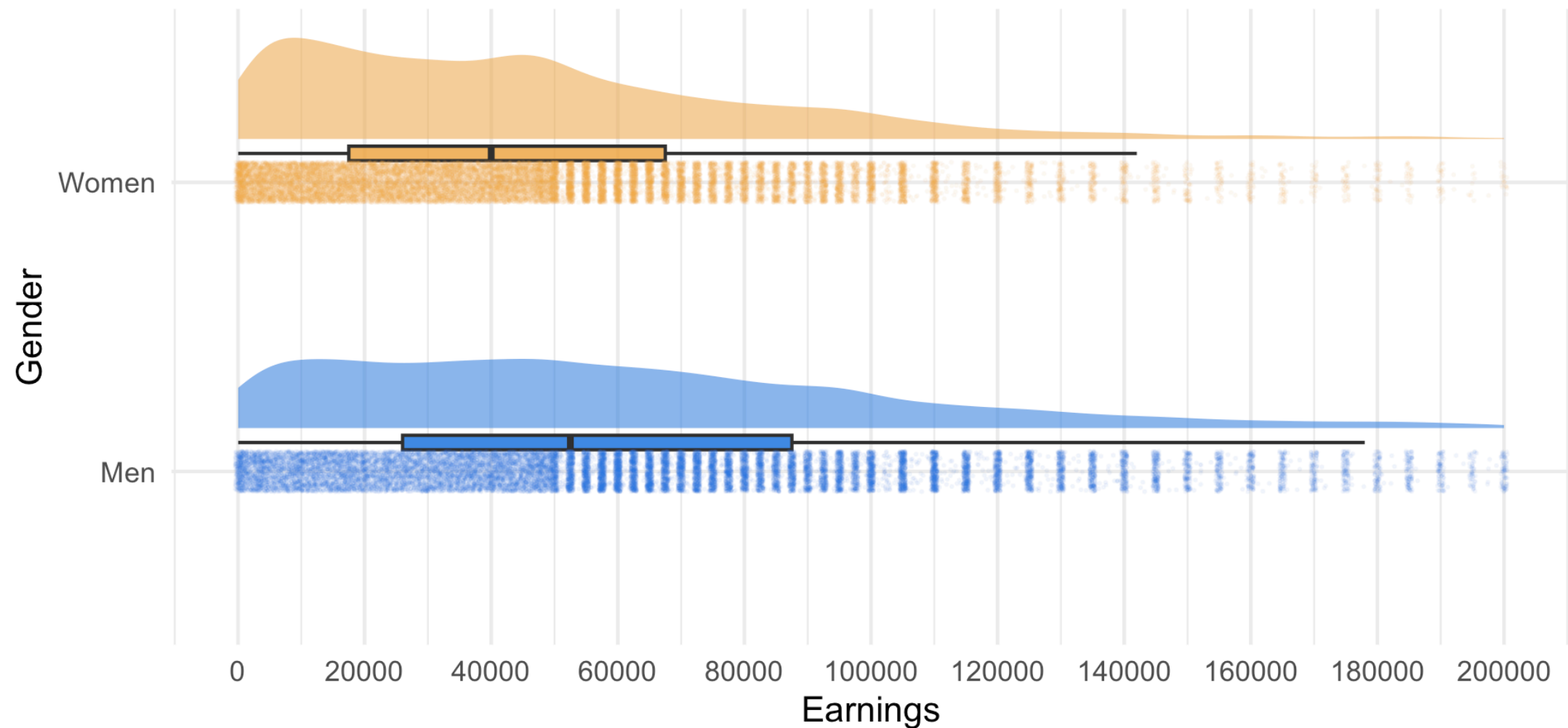
Source: Canadian Income Survey (2022). Plot includes wages from the \$5 to \$100

Earnings by Highest Level of Schooling



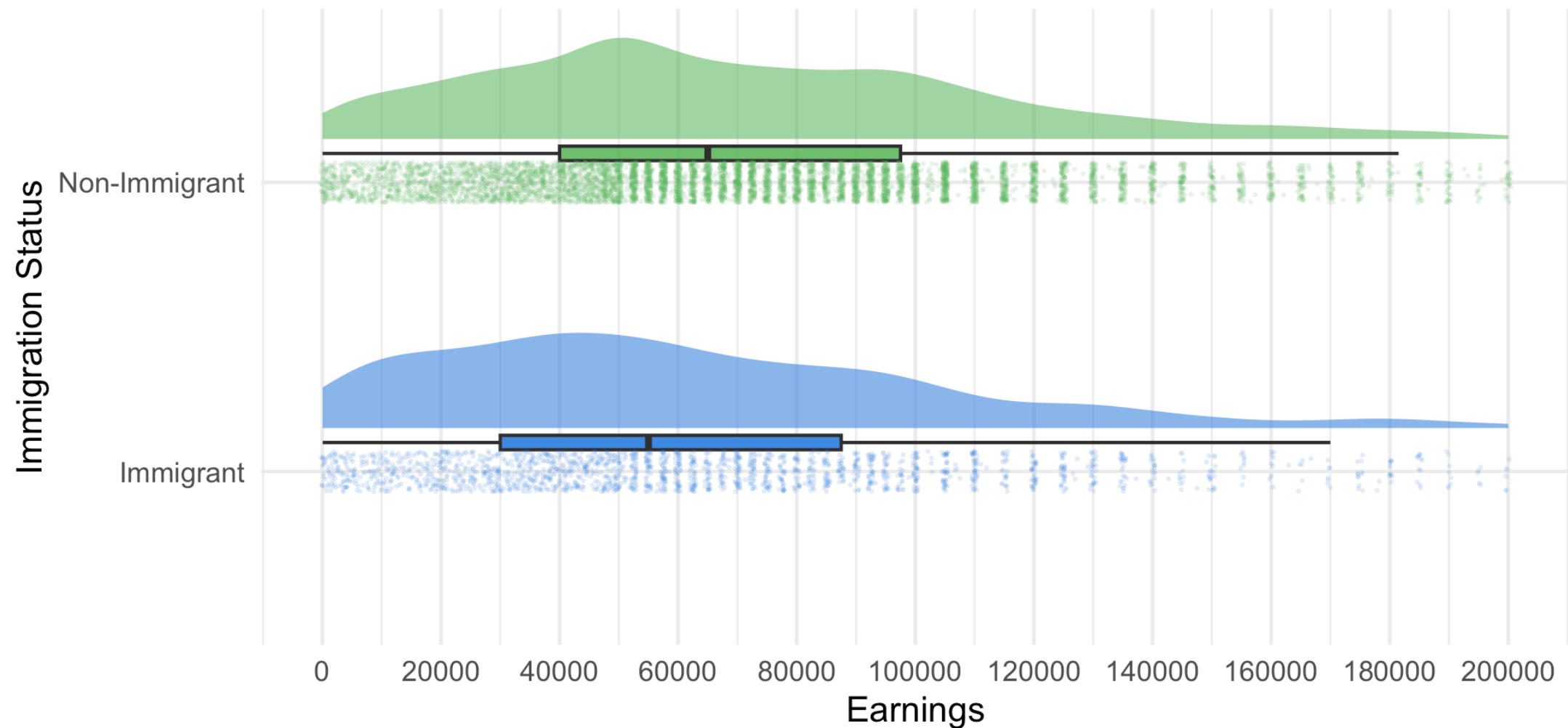
Source: Canadian Income Survey (2022). Plot includes individuals aged 40-50, earnings from the 1st percentile to the 98th percentile

Earnings by Gender



Source: Canadian Income Survey (2022). Plot includes earnings from the 1st percentile to the 98th percentile

Earnings by Immigration Status



Source: Canadian Income Survey (2022). Plot includes aged 40-50, earnings from the 1st percentile to the 98th percentile

iClicker Question

The charts show men earning more than women on average. Based on these data alone, we can conclude:

- A) Discrimination causes the gap
- B) Women choose lower-paying occupations
- C) The gap is real, but these data cannot tell us why
- D) The gap is fully explained by differences in hours worked
- E) There is no meaningful gap

Empirical Research

Correlation

! Key Definition

The **correlation coefficient** describes the relationship between two variables, ranging from -1 to 1 .

- Close to **1**: strong positive · 0: none · close to **-1**: strong negative
- Across countries, participation and log national income correlate at 0.28 for women and 0.41 for men

Regression Analysis

- Regression fits the line that **best fits** the data, rather than eyeballing it

$$\text{Participation rate} = a + b \times \ln(GNI) + e$$

- **Ordinary least squares** chooses the line minimizing the sum of squared residuals
- Estimating across 168 countries gives a slope of 4.12 for women
 - A one unit increase in log income is associated with a 4.12 percentage point higher participation rate
- **R-squared** measures goodness of fit — income explains only 7.85% of the variation in women's participation
- Every estimate comes with a **standard error** measuring its precision
 - Rule of thumb: a coefficient at least **twice its standard error** is **statistically significant** — unlikely to be zero

Multiple Regression

- We can add more variables. For the union wage differential:

$$\text{Weekly Earnings} = a + b \text{ Unionized} + e$$

- Adding controls for education, gender, age, and industry shrinks the differential
- Adding **job tenure** shrinks it further — even negative in some provinces

Warning

But if a main role of unions is **job security** — letting people stay in higher-paying jobs — then controlling for tenure understates how much unions matter. Be careful what you control for.

Correlation versus Causation

- The number of movies Margot Robbie appears in correlates with the number of firefighters in South Dakota
- The divorce rate in Maine correlates with margarine consumption in the United States

! Key Definition

These are **spurious correlations**. A strong correlation does not mean one variable is **causing** changes in the other.

- We usually want **causal** relationships, not just associations — which is what the next section is about

iClicker Question

Adding job tenure as a control shrinks the estimated union wage differential. Why should we be cautious?

- A) Job tenure is measured with error
- B) If unions raise wages partly by increasing job security, controlling for tenure removes part of the effect
- C) Job tenure is unrelated to earnings
- D) Adding more controls always biases estimates downward
- E) The sample becomes too small

Advanced Research Design in Labour Economics

The Fundamental Problem

- University graduates earn more (Chapter 14) — but is that **causal**?
 - Perhaps they already had marketable skills that would have raised their wages anyway
 - We suspect **confounding factors** driving both wages and schooling

! Key Definition

The **counterfactual outcome** is what would have happened to the treated had they *not* received the intervention. We can never observe it.

The Fundamental Problem

You injure your ankle, your doctor prescribes physiotherapy, and six weeks later it is healed.

Did the physiotherapy **cause** it to heal? Or would it have healed anyway?

- You cannot answer this, because you cannot observe the healing that would have happened without treatment
- **Research designs** are strategies for estimating that missing counterfactual

Why bother?

To design policy that addresses causes rather than symptoms, to predict what happens when conditions change, and to anticipate **unintended consequences**.

Experimental Methods

Key Definition

Random assignment to treatment and control groups. The gold standard.

- We simply compare the mean outcomes of the two groups
- Easy to explain to, and be understood by, policymakers and the public
- Everything else is **nonexperimental**, using observational data

Tip

In 2021 the Canadian labour economist **David Card** won the Nobel Prize for his empirical contributions to labour economics, sharing it with Joshua Angrist and Guido Imbens for their work on analyzing causal relationships.

Audit and Résumé Studies

Key Definition

Audit studies approximate random assignment by comparing outcomes where the treatment is manipulated or hidden.

- Résumé studies submit artificially manipulated resumés to real job ads and record which get **call backs**
- Resumés are varied to signal demographic differences — common foreign versus domestic names, volunteer work signalling child care responsibilities, activities signalling sexual orientation

Several examples appear in Chapter 15.

Simple Comparisons and Their Problems

Before-and-after comparison of those treated

- Compares the treated before and after; needs longitudinal data
- Assumes pre-treatment outcomes estimate the counterfactual — often **unconvincing**
 - Some unemployed workers would have found jobs without a job search program

Post-treatment comparison of treated and comparison groups

- Compares the treated to a similar untreated group; needs only cross-section data
- But with no pre-treatment data, **existing differences** between the groups bias the estimate

Difference-in-Differences

Key Definition

Take both groups, **before and after**. Contrast each group's change over time; the **difference in those two differences** estimates the causal effect.

- This fixes the main limitations of both simpler designs
- The comparison group's change over time estimates the counterfactual

Tip

We use this twice: Maine vs New Brunswick in Chapter 5, and Card and Krueger in Chapter 12.

Selection Bias and Natural Experiments

The main threat

Selection bias — unobserved factors that determine who gets treated *and* affect the outcome. Regression cannot control for what you cannot see.

- Training program participants chose to apply, perhaps because of motivation the researcher never observes

Key Definition

A **natural experiment** arises when an event — a policy change, a disaster — affects some people and not others, for reasons **unrelated** to the treatment of interest.

- Because nobody chose to be affected, selection bias is much less of a problem

Discontinuities and Instruments

! Regression discontinuity design

When treatment depends on a **score** crossing a threshold, compare those just above and just below. A student scoring 92 and one scoring 91 are otherwise **nearly identical** — the jump at the threshold estimates the effect.

! Instrumental variables

Find an **instrument** that shifts the treatment exogenously but does not affect the outcome directly — the **exclusion restriction**. Compulsory schooling laws instrument for education in Chapter 14.

Building a Comparison Group

! Fixed effects

Panel data let us use only the changes a *given individual* makes, differencing out anything fixed within them — motivation, ability. The twin studies in Chapter 14 work this way.

! Matching and propensity scores

Construct a control group that looks like the treated. The **propensity score** is each person's estimated probability of being treated; match treated and untreated people with the same score.

! Synthetic comparison groups

With one treated group and several imperfect candidates, build a weighted comparison group matching the treated group's pre-treatment path.

Research Designs: Summary

Design	Main worry
Audit / resumé studies	External validity
Before-and-after	No counterfactual
Post-treatment comparison	Pre-existing differences
Difference-in-differences	Selection bias
Natural experiment	Is the event really exogenous?
Regression discontinuity	Effects only near the cut-off
Instrumental variables	Exclusion restriction
Fixed effects	Only handles <i>fixed</i> unobservables
Matching / propensity score	Unobservables remain
Synthetic comparison	Pre-treatment fit

iClicker Question

You have earnings before and after a new training program, for participants and for a comparable group of non-participants. Which design does this support?

- A) A randomized experiment
- B) Before-and-after comparison of those treated
- C) Difference-in-differences
- D) A regression discontinuity design
- E) Instrumental variables

Summary

Summary

- Participation varies enormously across countries, and most of that variation is in [women's](#) participation
- Which employment law applies to you depends on your employer's [industry](#), not where you live
- The LFS, Census, and CIS are the workhorses; the LFS defines every unemployment number you hear
- A regression coefficient is not automatically a [causal effect](#)
- Estimating a causal effect means estimating a [counterfactual](#) we can never observe — every research design is a different way of building one

When you see evidence in this course, ask what the [comparison group](#) is, and whether you believe it.